

数据库实验报告（第四周）

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一、实验目的

- 熟悉SQL的数据查询语言
- 能够使用SQL语句对数据库进行嵌套查询、集合运算

二、实验内容

1. 查询选修C++课程的成绩比姓名为 ZNKOO的学生高的所有学生的编号和姓名
2. 找出和学生883794999或学生850955252的年级一样的学生的姓名
3. 查询没有选修Java的学生名称
4. 找出课时最少的课程的详细信息
5. 查询工资最高的教师的编号和开设的课程号
6. 找出选修课程ERP成绩最高的学生编号
7. 查询没有学生选修的课程名称
8. 查询讲授课程UML的教师所讲授的所有课程名称
9. 使用集合交运算，查询既选修了database又选修了UML课程的学生编号
10. 使用集合减运算，查询选修了database却没有选修UML课程的学生编号

说明：所有查询均在 `USE School_Data;` 上下文下执行。

三、实验结果

(1) 查询选修C++课程的成绩比姓名为 ZNK00的学生高的所有学生的编号和姓名

```
SELECT DISTINCT
    STUDENTS.sid AS 学生编号,
    STUDENTS.sname AS 学生姓名
FROM
    STUDENTS
    JOIN CHOICES ON CHOICES.sid = STUDENTS.sid
    JOIN COURSES ON COURSES.cid = CHOICES.cid
WHERE
    COURSES.cname = 'C++'
    AND STUDENTS.sname != 'ZNK00'
    AND CHOICES.score > (
        SELECT score
        FROM
            STUDENTS s
            JOIN CHOICES c ON s.sid = c.sid
            JOIN COURSES co ON c.cid = co.cid
        WHERE
            s.sname = 'ZNK00'
            AND co.cname = 'C++'
    );
```

```
3  -- (1)查询选修C++课程的成绩比姓名为 ZNK00的学生高的所有学生的编号和姓名;
4  select distinct
5      STUDENTS.sid,
6      STUDENTS.sname
7  from
8      STUDENTS
9      join CHOICES 1<->1..n: on CHOICES.sid = STUDENTS.sid
10     join COURSES 1..n<->1: on COURSES.cid = CHOICES.cid
11  where
12      COURSES.cname = 'C++'
13      and STUDENTS.sname != 'ZNK00'
14      and CHOICES.score > (
15          select score
16          from
17              STUDENTS s
18              join CHOICES c 1<->1..n: on s.sid = c.sid
19              join COURSES co 1..n<->1: on c.cid = co.cid
20          where
21              s.sname = 'ZNK00'
22              and co.cname = 'C++'
23      )
24  )
```

输出 (1)查询选修C++课程的成绩比姓名为 ...ZNK00的学生高的所有学生的编号和姓名; × (2)找出和学生883

4

	sid	sname
1	849943768	vteben
2	850129624	nvibxo
3	850270642	cjnzj
4	850352634	qqdvduyn
5	850480913	fghwa
6	851056920	mhaafbyvi
7	851087606	wwwqvcor
8	851406411	nda1danen

2,048行

(2) 找出和学生883794999或学生850955252的年级一样的学生的姓名

```
SELECT DISTINCT
    sname AS 学生姓名
FROM STUDENTS
WHERE
    STUDENTS.sid != '883794999'
    AND STUDENTS.grade IN (
        SELECT grade
        FROM STUDENTS
        WHERE
            STUDENTS.sid = '883794999'
    )
UNION
SELECT DISTINCT
    sname AS 学生姓名
FROM STUDENTS
WHERE
    STUDENTS.sid != '850955252'
    AND STUDENTS.grade IN (
        SELECT grade
        FROM STUDENTS
        WHERE
            STUDENTS.sid = '850955252'
    );
```

这里我将这两个学生本人排除在外，如果加上，则行数应为 $13369+2=13371$

```
select distinct
    sname
from STUDENTS
where
    STUDENTS.sid != '883794999'
    and STUDENTS.grade in (
        select grade
        from STUDENTS
        where
            STUDENTS.sid = '883794999'
    )
union
select distinct
    sname sname
from STUDENTS
where
    STUDENTS.sid != '850955252'
    and STUDENTS.grade in (
        select grade
        from STUDENTS
        where
            STUDENTS.sid = '850955252'
```

输出 (1)查询选修C++课程的成绩比姓名为 ...ZNKOO的学生高的所有学生的编号和姓名; (2)找出和学生883794999或学生...850955252的年级一样的学生的姓名; ×

	sname
1	aaaadsg
2	aabskm
3	aacef
4	aactnzls
5	aactzdn
6	aagendsis
7	aagtfy
8	aahkilft

13,369 行

(3) 查询没有选修Java的学生名称

```
SELECT SNAME
FROM STUDENTS
WHERE NOT EXISTS(
SELECT *
FROM CHOICES
JOIN COURSES ON CHOICES.CID=COURSES.CID
WHERE STUDENTS.SID=CHOICES.SID
AND COURSES.CNAME='Java')
```

-- (3) 查询没有选修Java的学生名称;

```
SELECT SNAME
FROM STUDENTS
WHERE NOT EXISTS(
SELECT *
FROM CHOICES
JOIN COURSES 1..n<->1: ON CHOICES.CID=COURSES.CID
WHERE STUDENTS.SID=CHOICES.SID
AND COURSES.CNAME='Java');
```

-- (4) 找出课时最少的课程的详细信息;

```
select *
```

ZNKOO的学生高的所有学生的编号和姓名;

(2) 找出和学生883794999或学生...850955252的年级一样的学生的姓名;



SNAME

1	frroly
2	mgllhxm
3	affmpv
4	drofja
5	cypxi
6	ogieznmh
7	wfyceta
8	mkcqigsyo

94,048 行

(4) 找出课时最少的课程的详细信息

```
SELECT
    col.cid AS 课程编号,
    col.cname AS 课程名称,
    col.hour AS 课时
FROM COURSES AS col
WHERE
    col.hour <= ALL (
        SELECT hour
        FROM COURSES
        WHERE
            COURSES.hour IS NOT NULL
    );
```

```

54  -- (4)找出课时最少的课程的详细信息;
55  select *
56  from COURSES as col
57  where
58      col.hour <= ALL (
59          select hour
60          from COURSES
61          where
62              COURSES.hour is not null
63      )
64

```

52的年级一样的学生的姓名; (3)查询没有选修Java的学生名称; (4)找出课时最少的课程的详细信息; ×

	cid	cname	hour
1	10024	use case	18
2	10034	windows	18

2 行

(5) 查询工资最高的教师的编号和开设的课程号

```

SELECT
    t.tid AS 教师编号,
    t.tname AS 教师姓名,
    ch.cid AS 课程编号,
    t.salary AS 工资
FROM
    TEACHERS AS t
    JOIN CHOICES AS ch ON t.tid = ch.tid
    JOIN COURSES AS co ON ch.cid = co.cid
WHERE
    t.salary >= ALL (
        SELECT salary
        FROM TEACHERS
        WHERE
            salary IS NOT NULL
    );

```

```

76  -- (5)查询工资最高的教师的编号和开设的课程号;
77  select t.tid, t.tname, ch.cid, t.salary
78  from
79      TEACHERS as t
80      join CHOICES as ch 1<->1..n: on t.tid = ch.tid
81      join COURSES as co 1..n<->1: on ch.cid = co.cid
82  where
83      t.salary >= ALL (
84          select salary
85          from TEACHERS
86          where
87              salary is not NULL
88      )

```

					; (3)查询没有选修Java的学生名称; (4)找出课时最少的课程的详细信息; (5)查询工资最高的教师的编号和开设的课程号; ×				
					Tx: 自动 DDL 84行				
4 4	1	277877392	wbvmggkvi	10041					4999
1	2	204711560	xarfnjdjtd	10001					4999
2	3	204711560	xarfnjdjtd	10026					4999
3	4	287866460	qzftezkyu	10005					4999
	5	277877392	wbvmggkvi	10047					4999
	6	214445507	eghgugq	10022					4999
	7	214445507	eghgugq	10050					4999
	8	204711560	xarfnjdjtd	10041					4999

(6) 找出选修课程ERP成绩最高的学生编号

```

SELECT DISTINCT
    ch.sid AS 学生编号
FROM CHOICES AS ch
    JOIN COURSES AS co ON ch.cid = co.cid
WHERE
    co.cname = 'ERP'
    AND ch.score >= ALL (
        SELECT score
        FROM CHOICES c
            JOIN COURSES co2 ON c.cid = co2.cid
        WHERE
            co2.cname = 'ERP'
            AND score IS NOT NULL
    );

```



```

90  -- (6)找出选修课程ERP成绩最高的学生编号;
91  select distinct
92      ch.sid
93  from CHOICES as ch
94      join COURSES as co  1..n<->1: on ch.cid = co.cid
95  where
96      co.cname = 'ERP'
97      and ch.score >= ALL (
98          select score
99          from CHOICES c
100             join COURSES co2  1..n<->1: on c.cid = co2.cid
101      where
102          co2.cname = 'ERP'
103          and score is not null
104      )
105

```

📊 (4)找出课时最少的课程的详细信息;

📊 (5)查询工资最高的教师的编号和开设的课程号;

📊 📈 🔄 ⌚ 🗄 + - ↶ ↷ ⬆ Tx: 自动 ▾ DDL 📌 🔍 🗨 📧

🔗 sid 🔍 ▾

1	800229512
2	800758618
3	802056337
4	803963938
5	803973094
6	804483690
7	804685587
8	808410625

95 行 ▾ | ⋮

(7) 查询没有学生选修的课程名称

```
SELECT DISTINCT
    cname AS 课程名称
FROM COURSES
EXCEPT
SELECT DISTINCT
    cname AS 课程名称
FROM CHOICES AS ch
    JOIN COURSES AS co ON ch.cid = co.cid;
```

```
SELECT cname AS 课程名称
FROM COURSES AS c
WHERE NOT EXISTS (
    SELECT 1
    FROM CHOICES AS ch
    WHERE ch.cid = c.cid
);
```

```
106 -- (7)查询没有学生选修的课程名称;
107 select distinct
108     cname
109 from COURSES
110 except
111 select distinct
112     cname cname
113 from CHOICES as ch
114     join COURSES as co 1..n<->1: on ch.cid = co.cid
115
```

号; (6)找出选修课程ERP成绩最高的学生编号; (7)查询没有学生选修的课程名称; ×

        

cname 

4 4
1
2
3

0行 ∨ ⋮

(8) 查询讲授课程UML的教师所讲授的所有课程名称

```
SELECT DISTINCT
  cname AS 课程名称
FROM
  TEACHERS AS t
  JOIN CHOICES AS ch ON ch.tid = t.tid
  JOIN COURSES AS co ON co.cid = ch.cid
WHERE
  t.tid IN (
    SELECT t.tid
    FROM
      TEACHERS AS t
      JOIN CHOICES AS ch ON ch.tid = t.tid
      JOIN COURSES AS co ON co.cid = ch.cid
    WHERE
      co.cname = 'UML'
  );
```

```
16  -- (8)查询讲授课程UML的教师所讲授的所有课程名称;
17  select distinct
18      cname
19  from
20      TEACHERS as t
21      join CHOICES as ch 1<->1..n: on ch.tid = t.tid
22      join COURSES as co 1..n<->1: on co.cid = ch.cid
23  where
24      t.tid in (
25          select t.tid
26          from
27              TEACHERS as t
28              join CHOICES as ch 1<->1..n: on ch.tid = t.tid
29              join COURSES as co 1..n<->1: on co.cid = ch.cid
30          where
31              co.cname = 'UML'
32      )
```

号;		(6)找出选修课程ERP成绩最高的学生编号;	(7)查询没有学生选修的课程名称;	(8)查询讲授课程UML的教师所讲授的所有课程名称;	×
		Tx: 自动 DDL 50行 CSV			
cname					
1	algorithm				
2	architectonics				
3	artifical intelligence				
4	asp				
5	basic				
6	c				
7	c#				
8	...				

(9) 使用集合交运算，查询既选修了database又选修了UML课程的学生编号

```
SELECT DISTINCT
    STUDENTS.sid AS 学生编号
FROM
    STUDENTS
    JOIN CHOICES ON STUDENTS.sid = CHOICES.sid
    JOIN COURSES ON COURSES.cid = CHOICES.cid
WHERE
    cname = 'database'
INTERSECT
SELECT DISTINCT
    STUDENTS.sid AS 学生编号
FROM
    STUDENTS
    JOIN CHOICES ON STUDENTS.sid = CHOICES.sid
    JOIN COURSES ON COURSES.cid = CHOICES.cid
WHERE
    cname = 'UML';
```

```

134 -- (9)使用集合交运算, 查询既选修了database又选修了UML课程的学生编号;
135 select distinct
136     STUDENTS.sid
137 from
138     STUDENTS
139     join CHOICES 1<->1..n: on STUDENTS.sid = CHOICES.sid
140     join COURSES 1..n<->1: on COURSES.cid = CHOICES.cid
141 where
142     cname = 'database'
143 intersect
144 select distinct
145     sid STUDENTS.sid
146 from
147     STUDENTS
148     join CHOICES 1<->1..n: on STUDENTS.sid = CHOICES.sid
149     join COURSES 1..n<->1: on COURSES.cid = CHOICES.cid
150 where
151     cname = 'UML'

```

课程名称;		(9)使用集合交运算, 查询既选修了da...tabase又选修了UML课程的学生编号; ×	
sid			
1	825682435		
2	862299211		
3	815042468		
4	855208874		
5	819990159		
6	868953550		
7	840368713		
8	879779584		

303行

(10) 使用集合减运算，查询选修了database却没有选修UML课程的学生编号

```
SELECT DISTINCT
  STUDENTS.sid AS 学生编号
FROM
  STUDENTS
  JOIN CHOICES ON STUDENTS.sid = CHOICES.sid
  JOIN COURSES ON COURSES.cid = CHOICES.cid
WHERE
  cname = 'database'
EXCEPT
SELECT DISTINCT
  STUDENTS.sid AS 学生编号
FROM
  STUDENTS
  JOIN CHOICES ON STUDENTS.sid = CHOICES.sid
  JOIN COURSES ON COURSES.cid = CHOICES.cid
WHERE
  cname = 'UML';
```

153 -- (10)使用集合减运算， 查询选修了database却没有选修UML课程的学生编号；

154 select distinct

155 STUDENTS.sid

156 from

157 STUDENTS

158 join CHOICES 1<->1..n: on STUDENTS.sid = CHOICES.sid

159 join COURSES 1..n<->1: on COURSES.cid = CHOICES.cid

160 where

161 cname = 'database'

162 except

163 select distinct

164 sid STUDENTS.sid

165 from

166 STUDENTS

167 join CHOICES 1<->1..n: on STUDENTS.sid = CHOICES.sid

168 join COURSES 1..n<->1: on COURSES.cid = CHOICES.cid

169 where

170 cname = 'UML'

名称; (9)使用集合交运算， 查询既选修了da...atabase又选修了UML课程的学生编号; (10)使用集合减运算， 查询选修了da...abase却没有选修UML课程的学生编号;

CSV

↓

↑

↶

sid

le_4 4:

le_1

le_2

le_3

1 803870334

2 815496921

3 815855012

4 891800873

5 882471429

6 889351393

7 887332774

8 846810167

5,454行

四、问题&总结

- 第(1)题：嵌套查询的关键在于子查询要返回单值。这里用子查询获取ZNKOO学生的C++成绩，然后在外层查询中比较其他学生的成绩。注意要用sname而不是sid来匹配学生姓名。
- 第(3)题：使用EXCEPT集合减运算时，第一个查询应该是所有学生，而不是只查选过课的学生，否则会漏掉没选任何课的学生。
- 第(6)题：题目要求查询ERP课程成绩最高的学生，需要在子查询中也限定ERP课程，不能查所有课程的最高分。
- 第(8)题：使用IN谓词配合子查询，先找出讲授UML课程的教师编号，再查询这些教师讲授的所有课程。这是典型的相关子查询应用。
- 集合运算：UNION（并）、INTERSECT（交）、EXCEPT（减）三种集合运算要求参与运算的两个查询结果必须有相同的列数和兼容的数据类型。使用时要注意列的对应关系。
- ALL谓词：在第(4)(5)(6)题中使用了 $\geq$ ALL 或 $\leq$ ALL 来查找最大值或最小值，这种方式比使用MAX/MIN函数更灵活，可以返回完整的记录信息。